5.2 Construct truth tables for logic statements

Truth Tables for Conditional Statements

Recall that a conditional statement takes the form $p \rightarrow q$.

A conditional statement is false only when p is true and q is false.

Consider the statement: “If you study for the test, then you will pass."

p: You study for the test.

q: You pass the test.

p	q	$p \rightarrow q$	Explanation
T	T	T	You studied and passed.
T	F	F	You studied but failed.
F	T	T	You didn't study but passed anyway.
F	F	T	You didn't study and failed.

Truth Tables for Biconditional Statements

Recall that a biconditional statement takes the form $p \leftrightarrow q$.

A biconditional statement is true when p and q have the same truth value (both true or both false).

Consider the statement: "You pass the test if and only if you get a grade of 50% or higher."

p: You pass the test.

q: You get a grade of 50% or higher.

p	q	$p \leftrightarrow q$	Explanation
T	T	T	You pass and receive at least 50%.
T	F	F	You pass but receive less than 50%.
F	T	F	You fail but received at least 50%.
F	F	T	You fail and receive less than 50%.

Truth Tables for Inverse Statements

Recall that the inverse of a conditional statement is formed by negating both statements, $\sim p \rightarrow \sim q$

Construct a truth table for $\sim p \rightarrow \sim q$.

p	q	$\sim p$	$\sim q$	$\sim p \rightarrow \sim q$

The inverse is a conditional statement, so the same rule applies; it is false only when the first part ($\sim p$) is true and the second part ($\sim q$) is false.

Examples

Construct a truth table for each of the following compound statements. Include columns for all intermediate operations.

1. $(p \wedge q) \rightarrow \sim p$

2. $(\sim p \vee q) \leftrightarrow p$

3. $(p \wedge q) \rightarrow r$

4. $(p \vee \sim q) \wedge r$

5. $(p \rightarrow q) \leftrightarrow \sim r$

6. $(\sim p \rightarrow \sim q) \vee (q \wedge r)$

5.3 Write algorithms using conditional logic to solve a problem

The logical conditions used in algorithms follow the same rules as the statements evaluated in truth tables. The result of a condition determines which action the algorithm performs.

An algorithm is a set of step-by-step instructions used to solve a problem.

In computer programming, algorithms often use conditional logic to make decisions based on whether a condition is true or false.

Conditional logic uses statements such as IF, THEN, and ELSE.

Pseudocode is a way of describing the steps of an algorithm using simple, English-like instructions rather than a specific programming language.

Command	**Purpose**
START	Begins the algorithm
INPUT	Accepts information
IF...THEN	Checks whether a condition is true
ELSE	Specifies what happens if the condition is false
OUTPUT	Displays a result
END IF	Marks the end of the conditional statement
END	Ends the algorithm

Examples

Write an algorithm using pseudocode to solve each of the following problems.

1. If a student's grade is at least 50%, then the student passes. Otherwise, the student fails.

```
START
   INPUT grade

   IF grade >= 50 THEN
      OUTPUT "Pass"
   ELSE
      OUTPUT "Fail"
   END IF
END
```

2. A customer qualifies for a discount if they are at least 65 years old AND have a membership card.

```
START
   INPUT age
   INPUT membership

   IF age >= 65 AND membership = “Yes” THEN
      OUTPUT "Discount applies"
   ELSE
      OUTPUT "No discount"
   END IF
END
```

Both conditions must be true for the discount to apply.

3. A student is eligible to write a supplementary exam if their final grade is between 45% and 49%, inclusive, OR they have special permission.

```
START
   INPUT grade
   INPUT permission

   IF (grade >= 45 AND grade <= 49) OR permission = “Yes” THEN
      OUTPUT "eligible"
   ELSE
      OUTPUT "not eligible"
   END IF
END
```

Note

Parentheses group the conditions that must both be true. A student qualifies if their grade is between 45% and 49%, inclusive, or if they have special permission.

Practice

Write an algorithm using pseudocode to solve each of the following problems.

1. Write an algorithm that accepts a temperature in degrees Celsius. If the temperature is 0°C or below, display "Freezing or Below". Otherwise, display "Above Freezing".

2. A company awards a bonus to employees who have worked for at least 5 years AND have received a performance rating of 4 or higher. Write an algorithm that determines whether an employee qualifies for the bonus.

3. An online store offers free shipping if a customer's order total is at least $75 OR the customer is a premium member. Write an algorithm that determines whether a customer qualifies for free shipping.

4. A user can access a computer system only if their account is NOT locked. Write an algorithm that determines whether access should be granted or denied.